NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND
MIL-PRF-24647B
2 April 1991
SUPERSEDING
DOD-P-24647A
23 October 1986
(See 6.7)

PERFORMANCE SPECIFICATION

PAINT SYSTEM, ANTICORROSIVE AND ANTIFOULING, SHIP HULL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ship hull anticorrosive and antifouling paint systems.

1.2 Classification. Paints covered by this specification are of the following types, classes, grades, and applications, as specified (see 6.2):

Type I - Paint systems having topcoats of ablative antifouling paints.
Type II - Paint systems having topcoats of non-ablative antifouling paints.
Class 1A - Paint systems having copper based toxics in the antifouling topcoats. Paint systems for use on metal substrates, except aluminum alloys.
Class 1B - Paint systems having mixed-toxics in the antifouling topcoats. Paint systems for use on metal substrates, except aluminum alloys.
Class 1C - Paint systems having antifouling topcoats that are toxic-free. Paint systems for use on all metal substrates, including aluminum.
Class 2 - Paint systems use on aluminum.
Class 3A - Paint systems having antifouling topcoats containing only copper-based toxics for use on rubber.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8010

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Class 3B - Paint systems having antifouling topcoats containing mixed-toxics for use on rubber.
Class 3C - Paint systems having antifouling topcoats that are toxic-free for use on rubber.
Grade A - Volatile organic content (VOC) of the antifouling topcoats shall be a maximum of 3.4 pounds per gallon [lb/gal; 400 grams per liter (g/L)]. VOC of all other individual paints in the paint system shall be in accordance with the requirements of Grade B.
Grade B - VOC of each individual paint of the paint system shall be a maximum of 2.8 lb/gal (340 g/L).
Grade C - VOC of each individual paint of the paint system shall be a maximum of 2.3 lb/gal (275 g/L).
Grade D - VOC of each individual paint of the paint system shall be a maximum of 0 lb/gal (0 g/L).
Application 1 - Paint systems having red color antifouling for underwater hull, with a service life of five years.
Application 2 - Paint systems having black color antifouling for waterline (boottop) of hull, with a service life of five years.
Application 3 - Paint systems having red color antifouling for underwater hull, with a service life of seven years.
Application 4 - Paint systems having black color antifouling for waterline (boottop) of hull with a service life of seven years.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons).

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PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-P-15931 - Paint, Antifouling, Vinyl (Formulas No. 121 and No. 129).
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.
MIL-A-21412 - Anodes, Corrosion Preventive, Magnesium Alloy, Cast or Extruded Shapes with Cast-In Cores.
MIL-P-24441 - Paint, Epoxy-Polyamide General Specification for.

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing.
FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.
FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U. S. Environmental Protection Agency (EPA)

40 CFR CH. 1, Part 60, Appendix A, Method 24 - Determination of Volatile Matter Content, Water Content, Density, Volume Solids and Weight Solids of Surface Coatings.

40 CFR 150-180 - Federal Insecticide, Fungicide and Rodenticide Act (FIFRA).

(Application for copies should be directed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC or to the Organization listed above.)

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Occupational Health and Safety Administration (OSHA)

29 CFR, Parts 1910, 1915, 1917, 1918, 1926, and 1928 - Hazard Communication Act, Final Rule.

(Application for copies should be directed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC or to the organization listed above.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

Threshold Limit Values (TLVs) for Chemical Substances and Physical Agents in the Work Environment and Biological Exposure Indices.

(Application for copies should be addressed to the American Conference of Governmental Hygienists, 6500 Glenway Avenue, Building D7, Cincinnati, OH 45211)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 209 - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate. (DOD adopted)

D 562 - Standard Test Method for Consistency of Paints Using the Stormer Viscometer. (DOD adopted)

D 660 - Standard Test Method for Evaluating Degree of Checking of Exterior Paints. (DOD) adopted)

D 661 - Standard Test Method for Evaluating Degree of Cracking of Exterior Paints. (DOD adopted)

D 714 - Standard Test Method for Evaluating Degree of Blistering of Paints. (DOD adopted)

D 1141 - Standard Specification for Substitute Ocean Water. (DOD adopted)

D 1475 - Standard Test Method for Density of Paint, Varnish, Lacquer, and Related Products. (DOD adopted)

D 1729 - Standard Practice for Visual Evaluation of Color Differences of Opaque Materials. (DOD adopted)

D 2000 - Standard Classification System for Rubber Products in Automotive Applications. (DOD adopted)

D 2244 - Standard Method for Calculation of Color Differences from Instrumentally Measured Color Coordinates. (DOD adopted)

D 2369 - Standard Test Method for Volatile Content of Solvent-Reducible Paints. (DOD adopted)

D 2698 - Standard Method for Determination of the Pigment Content of Solvent-Reducible Paints by High-Speed Centrifuging. (DOD adopted)

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D 3278 - Standard Test Methods for Flash Point of Liquids by Setaflash Closed-Cup Apparatus. (DOD adopted)
D 3623 - Standard Method for Testing Antifouling Panels in Shallow Submergence.
D 4400 - Standard Test Method for Sag Resistance of Paints Using a Multinotch Applicator.
D 4939 - Standard Test Method for Subjecting Marine Antifouling Coating to Biofouling and Fluid Shear Forces in Natural Seawater.
E 97 - Standard Test Method for Directional Reflectance Factor, 45-Deg 0-Deg, of Opaque Specimens by Broad-Band Filter Reflectometry. (DOD adopted)
F 718 - Standard for Shipbuilders and Marine Paints and Coating Product/Procedure Data Sheet. (DOD adopted)
F 1133 - Standard Practice for Inspecting the Coating System of a Ships' Underwater Hull and Boottop during Drydocking.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The paints furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.4).

3.2 Description of paints. Each paint system being qualified to a type, class, grade and application of this document shall consist of the following individual paints:

a. Anticorrosive paint (primer) intended for direct application to the substrate being painted.

b. Anticorrosive topcoat(s) intended for application over a primer. The topcoat of the anticorrosive layer shall be provided in FED-STD-595 color number 26260 (Haze gray), $R_0$1.8, $R_0$3.6 and $R_0$6 dark gray colors (see 3.4.1).

c. Tie coat(s), if required, intended for application to an anticorrosive paint and overcoated with antifouling paint.

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d. Type I paint systems shall have one of the following as the first coating of the total ablative antifouling topcoat system.

- a non-ablative commercial designation antifouling paint which has been qualified by the manufacturer to MIL-P-15931.

- a non-ablative antifouling paint qualified to type II of this document.

e. Antifouling topcoat(s).

Paint systems submitted for qualification to classes 1 and 2 of this specification are required to be qualified with anticorrosive paints to MIL-P-24441 and may, in addition, be qualified with a commercial product by the same manufacturer, used either directly or with an adhesion promoting coating (tie coat). Paint systems submitted for qualification to class 3 (rubber use) shall not need to qualify with anticorrosive coatings, but must be qualified with any required tie coat. Manufacturer shall at the time of qualification submittal identify all paints constituting the systems being qualified. At a minimum, the manufacturer shall identify the following unique nomenclature (product numerical and name designation, as well as color):

a. The product(s) constituting the antifouling paint topcoat(s).

b. Type I only - the commercial product designation for the type II non-ablative antifouling topcoat or MIL-P-15931 antifouling paint used under the type I coating.

c. All types, classes, and grades - any required tie coat.

d. Classes 1 and 2 - the commercial product designation for the vendor's approved MIL-P-24441 and commercial anticorrosive coatings being qualified with the antifouling paint.

3.3 Material Requirements. The paint systems furnished under this specification, when mixed and applied in accordance with the manufacturer's instructions, shall produce a coating that meets all the requirements of this specification. The composition of the paint furnished under this specification shall be the responsibility of the manufacturer, except as limited by this specification.

3.3.1 Types. Two types of antifouling topcoats which shall function in accordance with an identifiable toxic release mechanism are specified in 3.3.2.1 and 3.3.1.2.

3.3.1.1 Type I. Type I is a paint system with an ablative antifouling topcoat which shall dissolve, polish, erode, or reduce in thickness as the primary condition for performing its antifouling function. Its primary function shall be by the leaching process and may also exercise the option to be rejuvenated by mechanical underwater brushing.

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3.3.2 Classes.

3.3.2.1 Class 1.

3.3.2.1.1 Class 1A. The antifouling topcoat(s) in the Class 1A paint systems shall contain only copper compounds as the toxic agent(s). Class 1A paint systems are intended for use on non-aluminum metals (alloys of iron, nickel, or copper) substrates.

3.3.2.1.2 Class 1B. The antifouling topcoat(s) in the Class 1B paint systems may contain a mixture of toxic agents, one or more of which must be a copper compound. Class 1B paint systems are intended for use on non-aluminum metal (iron alloys, nickel alloys, copper alloys) substrates.

3.3.2.1.3 Class 1C. The antifouling topcoat(s) in the Class 1C paint systems shall not contain any toxic agent requiring EPA registration under FIFRA. Class 1C paint systems are intended for use on metal (iron alloys, aluminum alloys, nickel alloys, copper alloys) substrates and shall not contain ingredients which will cause aluminum corrosion.

3.3.2.2 Class 2. The antifouling topcoat(s) in the Class 2 paint systems may contain a mix of toxic agents. Class 2 paint systems are intended for use on aluminum alloys and shall be free of copper compounds or other ingredients corrosive to aluminum.

3.3.2.3 Class 3.

3.3.2.3.1 Class 3A. The antifouling topcoat(s) in the Class 3A paint systems shall contain only copper compounds as the toxic agent(s). Class 3A paint systems are intended for use on rubber substrates.

3.3.2.3.2 Class 3B. The antifouling topcoat(s) in the Class 3B paint systems may contain a mixture of toxic agents, one or more of which must be a copper compound. Class 3B paint systems are intended for use on rubber substrates.

3.3.2.3.3 Class 3C. The antifouling topcoat(s) in the Class 3C paint systems shall not contain toxic agents requiring EPA registration under FIFRA. Class 3C paint systems are intended for use on rubber substrates.

3.3.3 Grades.

3.3.3.1 Grade A. The VOC of antifouling topcoats of grade A paints shall not exceed 3.4 lb/gal (400 g/L) when in the ready-to-apply condition. The VOC of all other individual paints in the paint system shall be a maximum of 2.8 lb/gal (340 g/L). Ready-to-apply condition includes any volatile solvents added to prepare the paint for application (see 3.8).

3.3.3.2 Grade B. The VOC of each individual paint in the grade B paint system shall not exceed 2.8 lb/gal (340 g/L) when in the ready-to-apply condition (see 3.8).

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3.3.3.3 Grade C. The VOC of each individual paint in the grade C paint system shall not exceed 2.3 lb/gal (275 g/L) when in the ready-to-apply condition (see 3.8).

3.3.3.4 Grade D. Grade D paint system shall contain no VOC (see 3.8).

3.3.4 Applications.

3.3.4.1 Applications 1 and 2. Applications 1 and 2 paint systems shall have a minimum service life of 5 years without failure due to loss of adhesion, blistering, flaking, depletion by excessive ablation or loss of antifouling capability (excepting minor sliming and biofouling from the boottop to the light load line.).

3.3.4.2 Applications 3 and 4. Applications 3 and 4 paint systems shall have a minimum service life of 7 years without failure due to loss of adhesion, blistering, flaking, depletion due to excessive ablation or loss of antifouling capability (excepting sliming and biofouling from the boot top to the light load line).

3.3.5 Composition. The manufacturer is given his choice in the formulation of the paints described by this document, provided the end product conforms to all requirements of this specification. Soluble metals content and total metal content of each individual paint of the paint system shall not exceed the values listed in tables I and II when tested in accordance with 4.5.11. Copper content listed in tables I and II do not apply to antifouling topcoat paints or the total paint system of classes 1A, 1B, 2, 3A and 3B.

TABLE I. Soluble metals content.

Requirement	Maximum
Metals content (soluble), mg/L	
Antimony and/or its compounds	15
Arsenic and/or its compounds	5
Barium and/or its compounds (excluding barite)	100
Beryllium and/or its compounds	0.75
Cadmium and/or its compounds	1
Chromium (VI) compounds	5
Chromium and/or chromium (III) compounds	560
Cobalt and/or its compounds	80
Copper and/or its compounds	25
Fluoride salts	180
Lead and/or its compounds	5
Mercury and/or its compounds	0.02
Molybdenum and/or its compounds	350

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TABLE I. Soluble metals content - Continued.

Requirement	Maximum
Nickel and/or its compounds	20
Selenium and/or its compounds	1
Silver and/or its compounds	5
Thallium and/or its compounds	7
Vanadium and/or its compounds	24
Zinc and/or its compounds	250

TABLE II. Total metals content.

Requirement	Maximum
Metals content (total), weight percent	
Antimony and/or its compounds	0.05
Arsenic and/or its compounds	0.05
Barium and/or its compounds (excluding barite)	1.00
Beryllium and/or its compounds	0.0075
Cadmium and/or its compounds	0.01
Chromium (VI) compounds	0.05
Chromium and/or chromium (III) compounds	0.25
Cobalt and/or its compounds	0.80
Copper and/or its compounds	0.25
Fluoride salts	1.80
Lead and/or its compounds	0.10
Mercury and/or its compounds	0.002
Molybdenum and/or its compounds	0.35
Nickel and/or its compounds	0.20
Selenium and/or its compounds	0.01
Silver and/or its compounds	0.05
Thallium and/or its compounds	0.07
Vanadium and/or its compounds	0.24
Zinc and/or its compounds	0.50

3.3.5.1 Identification characteristics. Manufacturer shall disclose to the government the identification characteristics required in table III in accordance with the test method listed therein for each individual paint (mixed and components, as applicable) in the paint system. Paint identification characteristics listed in table III, when tested in accordance with requirements in 4.3, shall conform to the values disclosed by the manufacturer.

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Table III. Identification characteristics. 1/

Test	Applicable FED-STD-141 test method	Applicable ASTM test method
Pesticide (report manufacturing range or variation, +/- percent) 2/	---	---
Volatiles (report manufacturing range or variation, +/- percent)	---	D 2369
Nonvolatile vehicle (report manufacturing range or variation, +/- percent)	4053.1	---
Consistency, gram equivalent KU (report manufacturing range or variation, +/- KU)	---	D 562
Weight per unit volume (report manufacturing range or variation, +/- lb/gal)	---	D 1475
Drying to touch time (report - maximum not to exceed 2 hours at 20 degrees Celsius [°C], ASTM F 718 film thickness (68 degrees Fahrenheit [°F]))	4061.2	---
Drying hard time (report - maximum not to exceed 16 hours at 20°C [68°F], ASTM F 718 film thickness)	4061.2	---
Flash Point (report - minimum - to be not less than 21°C [70°] required)	---	D 3278
Volacile organic content (VOC) 3/	---	
Pot life (where applicable) (report hours)	---	---
Color $R_0$1.8, $R_0$3.6, and $R_0$6 dark gray colors 4/ All other paint system colors	--- 4250.1	--- ---
Infrared spectra of nonvolatile vehicle	---	D 2698
Sag resistance (report minimum value)	---	D 4400 (Method A)
Toxicity 5/		
Erosion or ablation rate (type I only 2/, 6/)		

1/ Manufacturing range percentages having variations of more than +/- 3 percent, without supporting documentation, will be rejected and can cause delay or rejection of qualification.
2/ Antifouling topcoats only. Report methods used to determine.
3/ VOC shall be determined in accordance with 4.5.3.
4/ Determine in accordance with 4.5.8.
5/ Certify in accordance with 3.3.5.2.
6/ Manufacturer shall establish and report erosion rate range in thousandths of an inch per year. This number shall be consistent with application approval sought and 4.5.1.

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3.3.5.2 Toxicity. Manufacturer shall prepare container label instructions for the paint in accordance with the OSHA requirements of 29 CFR, parts 1910, 1915, 1917, 1918, 1926, and 1928 (see 5.1). The paint shall contain no asbestos or asbestos form pigments, benzene, toluene, chlorinated solvents, hydrolyzable chlorine derivatives, coal tar or coal tar derivatives, lead compounds (not to exceed 0.06 percent of dry paint), chromate, cadmium, mercury or mercury compounds, or any ACGIH carcinogenic or ACGIH suspected carcinogenic compounds (see 3.5, 3.7, and 6.3).

3.3.5.3 Toxic products. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.4 Qualitative requirements.

3.4.1 Color. Colors of the individual paints of the paint systems shall:

a. When tested in accordance with 4.5.8.1, the colors of the individual anticorrosive and tie coat paints, except R_0 color paints, shall be in accordance with the FED-STD-595 color provided as an identification characteristic for that paint.

b. When tested in accordance with 4.5.8.2, the $R_0$1.8, $R_0$3.6 and $R_0$6 dark gray color anticorrosive topcoats of the paint system shall be as follows:

$R_0$1.8, measured as R_w, shall be R_w = 0.9 +/- 0.2.
$R_0$3.6, measured as R_w, shall be R_w = 1.9 +/- 0.2.
$R_0$6, measured as R_w, shall be R_w = 2.8 +/- 0.2.

c. The color of applications 1 and 3 antifouling topcoat paints shall be consistent with the pigments used. The color of application 2 and 4 antifouling topcoat paints shall be slightly darker than FED-STD-595 number 37056 when tested in accordance with 4.5.8.

If doubt exists as to whether the match in (a) or (c) is satisfactory, the color shall not exceed five color difference units in the direction of minus delta L when tested in accordance with table IV.

3.4.2 Condition in container. The coatings shall be readily mixed with a paddle or mechanical mixer to a smooth, uniform consistency after stirring for five minutes. Coatings shall not liver, hard settle, skin, seed or otherwise become unfit for use. Water based products shall not show phase separation, evidence of biological growth, putrefaction, corrosion of the container, or persistent foam which lasts more than 5 minutes. Test as specified in table IV.

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3.4.3 Storage stability. A previously unopened, original container of paint when stored and tested as required in 3.4.3.1 through 3.4.3.3 shall:

a. Readily mix with a paddle or mechanical mixer within five minutes to a smooth uniform condition.

b. Be free of grits, seed, tough or gummy sediment, skins, hard pigment settling, and persistent foam.

c. Not vary from the requirements established by table III by more than the following:

(1) ASTM D 562 consistency shall be not greater than five Kreb units (KU) higher or lower than table III identification characteristic value.

(2) Dry-to-touch and dry-hard times shall not be more than 60 minutes longer than the maximum specified in table III.

(3) Color shall be as specified in 3.4.1.

(4) Each individual paint of the paint system shall be readily applied to the manufacturer's ASTM F 718 wet film thickness with no sagging, running, or streaking.

3.4.3.1 Shelf life. An original, unopened, full, container of paint shall meet the requirements of 3.4.3 when tested as specified in 4.5.4.1.

3.4.3.2 Partially full container. Each individual paint or paint component shall meet the requirements of 3.4.3 when tested as specified in 4.5.4.2.

3.4.3.3 Accelerated storage stability. An unopened, original container of each individual paint shall meet the requirements of 3.4.3 when tested as specified in 4.5.4.3.

3.4.4 Resistance to tropical biofouling organism attachment (see 6.1.1). In all 3.4.4 tests, biofouling attributable to physical damage that has removed the antifouling topcoat paints shall be ignored in the evaluation of biofouling resistance performance.

3.4.4.1 Erosion.

3.4.4.1.1 Type I, all classes, all grades, applications 1 and 2 and type II, classes 1C and 3C, all grades, applications 1 and 2 paints shall meet the following requirements when tested in accordance with 4.5.1.3.1:

a. The anticorrosive paint layer shall not be exposed at any point due to erosion of the antifouling paint. The anticorrosive paint layers shall be free of peeling or flaking and shall not be rated: (1) Lower than 9 (1 percent) for checking, (2) lower than 9 for cracking, or (3) lower than 9 for blistering.

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b. No more than 10 percent of the type II or commercial designation MIL-P-15931 paint layer shall be exposed.

c. The antifouling topcoat shall not be rated: (1) lower than 9 (1 percent) for checking, (2) lower than 9 for cracking, and (3) lower than 10 for blistering.

d. The ablation or erosion rate of the antifouling topcoat shall be consistent with the erosion rate identification characteristic (see table III).

3.4.4.1.2 Type I, all classes, all grades, application 3 and 4 and type II, classes 1C and 3C, all grades, applications 3 and 4 paints shall meet the requirements of 3.4.4.1.1 when tested in accordance with 4.5.1.3.2.

3.4.4.2 Shallow submergence.

3.4.4.2.1 Types I and II, classes 1A, 1B, 2, 3A and 3B, all grades, applications 1 and 2 paints shall meet the following requirements when tested as specified in 4.5.1.4.1:

a. The anticorrosive paint layer shall not be exposed at any point due to erosion of the antifouling paint. The anticorrosive paint layers shall be free of peeling or flaking and shall not be rated: (1) lower than 9 (1 percent) for checking, (2) lower than 9 for cracking, or (3) lower than 9 for blistering.

b. The antifouling topcoat shall not be rated: (1) lower than 9 (1 percent) for checking, (2) lower than 9 for cracking, or (3) lower than 10 for blistering.

c. The biofouling resistance (FR) rating of the antifouling topcoat paint shall not be less than 100 (slime excluded) at any time during the test.

d. The antifouling film (AF) rating of the antifouling topcoat paint shall not be less than 100 at any time during the test.

e. The overall performance (OP) rating of the paint system shall not be less than 85 at any time during the test.

3.4.4.2.2 Types I and II, classes 1A, 1B, 2, 3A, and 3B, all grades, applications 3 and 4 paint systems shall meet the requirements of 3.4.4.2.1 when tested in accordance with 4.5.1.4.2.

3.4.4.3 Ship tests. Ship tests shall meet the requirements of 3.4.4.3.1 or 3.4.4.3.2, as appropriate as specified in 3.4.4.3.3 through 3.4.4.3.7.

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3.4.4.3.1 Type I, all classes, all grades, all applications paints shall meet the following requirements for a period equal to the service required for the appropriate application:

a. The antifouling topcoat shall have no biofouling of any kind, including slime, on at least 95 percent of the area painted.

b. Antifouling topcoats shall be free of film failure, such as peeling, flaking, softening, dissolving, and not wave eroded through to the underlying commercial designation MIL-P-15931 or type II coating. The antifouling paint shall not be rated: (1) lower than 9 (1 percent) for checking, (2) lower than 9 for cracking, and (3) less than 9 for blistering.

c. The erosion rate of the antifouling topcoat shall be consistent with the erosion rate identification characteristic (table III).

d. Anticorrosive paint layer shall not be rated: (1) lower than 9 (1 percent) for checking, (2) lower than 9 for cracking, and (3) less than 9 for blistering.

e. The paint systems shall have an OP rating for all factors of not less than 85 percent.

3.4.4.3.2 Type II, all classes, all grades, all applications paints shall meet the following requirements for the service period required for the appropriate application:

a. Classes 1A, 1B, 2, 3A, and 3B shall have no biofouling of any kind, except slime, on at least 95 percent of the area painted. Classes 1C and 3C shall be free of slime and shall have no biofouling on 85 percent of area painted.

b. The antifouling topcoats shall be free of film failure, such as peeling, flaking, softening, dissolving, and shall not have eroded through to the underlying anticorrosive coating. The antifouling paint shall not be rated: (1) lower than 9 (1 percent) for checking, (2) lower than 9 for cracking, and (3) less than 9 for blistering.

c. Anticorrosive paints shall be free of peeling and flaking and shall not be rated: (1) lower than 9 (1 percent) for checking, (2) lower than 9 for cracking, and (3) less than 9 for blistering.

d. The paint system shall have an OP rating for all factors of not less than 85 percent.

3.4.4.3.3 Commercial ship patch test. A test patch on a commercial ship, applied and tested as specified in 4.5.1.5.1 shall meet the requirements of 3.4.4.3.1 or 3.4.4.3.2, as appropriate.

3.4.4.3.4 Commercial ship test. A full ship application on a commercial ship, applied and tested as specified in 4.5.1.5.2 shall meet the requirements of 3.4.4.3.1 or 3.4.4.3.2, as appropriate.

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3.4.4.3.5 Government ship patch test. After completion of the second year of the tests required in 3.4.4.2, and/or 3.4.4.3.3, and/or 3.4.4.3.4, at the convenience of the Government, a ship patch of the proposed system applied to a government ship and tested as specified in 4.5.1.5.3 for a minimum period of one year shall meet the requirements of 3.4.4.3.1 or 3.4.4.3.2, as appropriate.

3.4.4.3.6 Government full ship test. After completion of the fourth year of the tests required in 3.4.4.2 and/or 3.4.4.3.3 and/or 3.4.4.3.4, at the convenience of the Government, a full ship application of the proposed system shall be applied to a government ship. Prior testing must include at least one year of the testing required by 3.4.4.2 and/or 3.4.4.3.3 and/or 3.4.4.3.4. The paint systems, when tested as specified in 4.5.1.5.4 for a minimum period of one year, shall meet the requirements of 3.4.4.3.1 or 3.4.4.3.2, as appropriate.

3.4.4.3.7 Service life determination. The Government, as specified in 4.5.1.5.5, will monitor the performance of qualified products on operational ships to establish compliance with the five year service requirement for application 1 and 2 and the seven year requirement for application 3 and 4 paint systems. The paint systems shall meet the requirements of 3.4.4.3.1 or 3.4.4.3.2, as appropriate.

3.4.5 Spraying properties. Each individual paint of the paint system, mixed in accordance with ASTM F 718, shall spray satisfactorily when tested as specified in 4.5.5. The dry film shall show no running, sagging, or streaking, blooming, blushing, bubbling, cratering, dusting, floating, fogging, hazing, mottling, orange peel appearance, pinholing, seeding, or streaking.

3.4.6 Brushing properties. Each individual paint of the paint system, mixed in accordance with ASTM F 718, shall brush easily and have good flowing and spreading qualities when tested as specified in 4.5.6. The dried film shall show no running, sagging, blooming, blushing, bubbling, cratering, dusting, floating, fogging, hazing, mottling, orange peel appearance, pinholing, seeding, or streaking.

3.4.7 Rolling properties. Each individual paint of the paint system, mixed in accordance with ASTM F 718 shall roll easily and have good flowing and spreading qualities when tested as specified in 4.5.7. The dried film shall show no running, sagging, blooming, blushing, bubbling, cratering, dusting, floating, fogging, hazing, mottling, orange peel appearance, pinholing, seeding, or streaking.

3.4.8 Recoatability. When tested as specified in 4.5.2, a recoat layer of antifouling paint shall show no peeling, flaking, or delamination from the overcoated antifouling paint layer. The recoat layer shall also meet all the requirements of 3.4.4.1.1 for type I paint and 3.4.4.1.2 for type II paint.

3.4.9 Cathodic protection (CP) compatibility. When tested as specified in 4.5.10, the test paint shall not peel, flake, blister, dissolve or otherwise fail. Lifting of the antifouling from the primer for 1/2 inch around the drilled hole shall be acceptable. This test shall not apply for coatings being qualified for class 3 (rubber substrates).

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3.4.10 Water based paints (ONLY). Water based (emulsion or water thinned) shall comply with the following provisions:

3.4.10.1 Resistance to biological growth. When tested as specified in table IV, the paint film shall have a surface disfigurement rating of 8 or greater.

3.4.10.2 Freeze-thaw stability. Each individual paint or paint component of the paint system shall meet the following requirements:

a. When opened, an original unopened container of each individual paint shall be free of putrification, phase separation, livering, curling, hard-dry caking, and tough gummy sediments.

b. Each individual paint or paint component of the paint system shall be capable of being mixed with a paddle or mechanical mixer within five minutes to a smooth, uniform state, free of gritty particles, putrification, phase separation, seeding, skins, pigment settling and persistent foam.

c. After mixing and testing in accordance with 4.5.9, each individual paint of the paint system shall be free of flocculation, gritty particles, putrification, phase separation, seeding, skins, pigment settling, persistent foam and the consistency shall not be more than plus or minus (+/-) 8 KU when compared to the consistency of the paint when originally received.

d. Each individual paint of the paint system shall meet the requirements of 3.4.3.

3.5 Directions for use. The manufacturer shall provide written directions on each container for the mixing and applying of the paint supplied and this direction shall include all information necessary to comply with the OSHA Hazard Communication Act and FED-STD-313. In addition, the manufacturer shall provide an ASTM F 718 data sheet which shall separately detail requirements for small unit containers (metric and english equivalents of pint, quart, gallon) and large unit containers (metric and english equivalents of five-gallon). Commercial and ASTM F 718 data sheets shall be included in each shipment of the paint covered by this specification (see 6.5).

3.6 EPA registration number. An U.S. EPA FIFRA antifouling paint registration number is required for all class 1A, 1B, 2, 3A, and 3B antifouling topcoat paints submitted for qualification to, or supplied to this specification. The container label shall be EPA approved (see 5.1). The manufacturer qualifying a class 1C and 3C antifouling topcoat paint may substitute a legal determination or EPA finding that the product does not require registration under the provisions of FIFRA .

3.7 Material safety data sheet (MSDS). The contractor shall prepare for each individual paint of the paint system a MSDS in accordance with the requirements of FED-STD-313 and OSHA Hazard Communication Act. The MSDS shall be included with each shipment of the material covered by this specification (see 6.5).

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3.8 Volatile organic content (VOC). The VOC of each individual paint in the paint system, when in the ready-to-apply condition, shall not exceed that allowed for the specific grade (see 3.3.3) for which qualification is being sought when tested in accordance with 4.5.3. Ready-to-apply condition includes any volatile solvents (thinners) required by the manufacturer's ASTM F 718 data sheets to prepare the paint for application.

3.9 Workmanship. The material shall be produced in accordance with the best commercial practices and shall be of a quality necessary to ensure a uniform, packaged product in conformance with the requirements of section 3 when test as specified in section 4. The paint described shall be suitable for the purpose intended (application to ship in dry-doc, under ambient condition, with conventional tools, and equipment).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that al products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall consist of all tests specified in table IV and paragraph 4.7, except 4.5.1.5.5, ship service life determination test. The products tested shall be materials drawn from regular production stocks. A test report shall be provided to the contracting officer or agent. Testing of products on Government ships shall be conducted at the convenience of the Government.

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TABLE IV. Test methods.

Requirements		Applicable FED-STD-141 test method	Applicable ASTM test method	Applicable MIL-P-24647 test method
*3.3.5 & TABLE I	Pigment (pesticide), percent	4021.1	---	---
*3.3.5 & TABLE I	Volatiles, percent	---	D 2369	---
*3.3.5 & TABLE I	Nonvolatile vehicle, percent	4053.1	---	---
*3.3.5 & TABLE I	Consistency, gram equivalent KU	---	D 562	---
*3.3.5 & TABLE I	Weight per unit volume, lb/gal	---	D 1475	---
3.3.5 & TABLE I	Copper content of pigment, percent	---	D 283	---
*3.3.5 & TABLE I	Drying time	4061.2	---	---
*3.3.5 & TABLE I	Flash point, °F	---	D 3278	---
3.3.3, 3.3.5, TABLE I, 3.8	Color	---		4.5.8
	Applications 1 and 3	---		4.5.8
	Applications 2 and 4 (Disputes only)	---		4.5.8
3.3.5 & TABLE I	Infrared spectra of nonvolatile vehicle	---	D 2698	---
*3.3.5 & TABLE I	Sag resistance (10 required)	---	D 4400 (method A)	---
3.3.5.2	Toxicity	---	---	3.3.5 and 3.5
3.3.3, 3.3.5 TABLE I	Lead content	---	D 3335	---
*3.4.2	Condition in container	3011.2	---	---
3.4.3.1	Shelf life	---	---	4.5.4.1
3.4.3.2	Partially full container	---	---	4.5.4.2
3.4.3.3	Accelerated storage stability	---	---	4.5.4.3
3.4.4	Resistance to tropical biofouling organism attachment	---	---	4.5.1
3.4.4.1	Erosion	---	D 4939	4.5.1.3
3.4.4.2	Shallow submergence	---	D 3623	4.5.1.4
3.4.4.3	Ship tests	---	---	4.5.1.5
*3.4.5	Spraying properties	---	---	4.5.5
3.4.6	Brushing properties	---	---	4.5.6
3.4.7	Rolling properties	---	---	4.5.7
3.4.8	Recoatability	---	---	4.5.2
3.4.9	Cathodic protection compatibility	---	---	4.5.10
3.4.10.2	Freeze-thaw stability	---		4.5.9

* See 4.4.

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the table IV tests marked with an asterisk (see 6.3). Tests shall be performed on each lot offered for delivery in accordance to the requirements of this specification.

4.4.1 Lot. A lot shall consist of an individual paint of the paint system of the same type, class, grade and application from a single uniform batch or a uniform blend of batches offered for delivery at one time.

4.4.1.1 Batch. A batch shall consist of a quantity of an individual paint of the paint system from a single finished product manufactured at the same time from the same units (sacks, cans, barrels, and so forth) or same supplier lot labeled ingredients.

4.4.2 Sampling. Two representative random samples from each lot shall be tested. The samples shall be packaged in separate containers. Minimum size for each sample shall be two containers of the size specified in the contract (see 6.2). If one or more defects are found in any sample, the entire lot shall be rejected. The contractor has the option of screening 100 percent of the rejected lot for the rejected characteristic(s), or providing a new lot which shall be inspected as specified in 4.4.

4.5 Test methods (all types). Test methods shall be as listed in table IV or as otherwise specified herein. The requirements listed in table IV shall be tested in accordance with the applicable tests and test paragraphs listed therein. Results of tests shall be in accordance with the requirements paragraph listed in table IV or in the test paragraph.

4.5.1 Resistance to tropical biofouling organism attachment.

4.5.1.1 Panel test requirements. Tests as specified in 4.5.1.3 and 4.5.1.4 shall be as follows:

a. Tests shall be conducted in Biscayne Bay, Florida; Daytona Beach, Florida; or Pearl Harbor, Hawaii.

b. Tests shall have six test panels prepared for each paint system required by ASTM D 3623 or D 4939, as appropriate.

c. Test panels shall be as follows: (1) Classes 1A, 1B, 1C test panels shall be made of the steel specified in ASTM D 4939; (2) Aluminum test panels for class 2 testing shall be ASTM B 209 (alloy 6061); (3) ASTM D 3623 test panels shall be a minimum of 1/8 inch thick; (4) Test panels for ASTM D 4939 shall be as required by ASTM D 4939; (5) Surface preparation of all steel and aluminum test panels shall be in accordance with ASTM D 4939 procedures; (6) Test panels for class 3 testing shall be of ASTM D 2000, type BC neoprene rubber having a Shore A Durometer 40 to 60 hardness and shall be at least 1/4 inch thick with surface preparation in accordance with the manufacturer's ASTM F 718 data sheet.

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d. Test panels shall be prepared for the qualifying systems and the control paint systems. Paint thicknesses of each individual coating and the total system shall be in accordance with the manufacturer's ASTM F 718 data sheet. Application of paint films shall be nominal in thickness (nominal is defined as +/- 25 percent of stated dry film thickness [DFT]). Rubber test panels do not require anticorrosive primer. Test systems consist of the following:

(1) Control (comparison) paint systems shall consist of ASTM D 4939, paragraph 7.1.2 paint system (utilizing the qualifying company's qualified type III and type IV formulas of the designated MIL-P-24441 anticorrosive) with the exception that the qualifying company's qualified MIL-P-15931 paint shall be the latest revision of that specification rather than the B revision cited.

(2) Test paint system no. 1 shall consist of the qualifying company's type III MIL-P-24441 formulas of the anticorrosive system specified in ASTM D 4939 materials and procedure Apply the manufacturer's ASTM F 718 antifouling system being qualified for use with this anticorrosion paint (see 3.3). Paints shall be applied and cured in accordance with the manufacturer's ASTM F 718 data sheet.

(3) Test paint system no. 2 shall consist of the qualifying company's type IV MIL-P-24441 formulas of the anticorrosive specified in ASTM D 4939. Apply the manufacturer's antifouling topcoat paint system (see 3.3). Paints shall be applied and cured in accordance with the requirements of the manufacturer's ASTM F 718 data sheet.

(4) Test paint system no. 3 shall consist of the manufacturer's commercial paint system and anticorrosive coatings, tie coats, if any, and antifouling paint(s) (see 3.3). Paints shall be applied and cured in accordance with the requirements of the manufacturer's ASTM F 718 data sheet.

e. Have erosion tests conducted in accordance with ASTM D 4939. An ASTM D 4939 cycle shall consist of 30 days static immersion followed by 30 days dynamic testing on a drum of 1.5 foot (nominal) diameter operated to provide a peripheral speed of 15+/-0.1 knots. Rate FR, AF, OP, and paint thickness in accordance with ASTM D 4939. Rate test films for: (1) checking, in accordance with ASTM D 660, (2) cracking, in accordance with ASTM D 661, and (3) blistering, in accordance with ASTM D 714. Inspect test samples for exposed type II, MIL-P-15931, or anticorrosion paint, as appropriate.

f. Have shallow submergence testing conducted in accordance with ASTM D 3623. Rate as follows: (1) FR, AF, and OP, in accordance ASTM D 3623, (2) checking, in accordance with ASTM D 660, (3) cracking, in accordance with ASTM D 661, and (4) blistering, in accordance with ASTM D 714. Inspect for exposed type II, MIL-P-15931, or anticorrosion paint, as appropriate.

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4.5.1.2 Ship test requirements. Ship tests (patch and full) required by 4.5.1.5 shall:

a. Have individual paints of the total system applied and cured in accordance with the requirements of the manufacturer's ASTM F 718 data sheet for those products (see 3.3).

b. Have the test ship(s) operate in high biofouling tropical or sub-tropical areas such as the Caribbean, Mediterranean or Western Pacific. Ship antifouling coatings shall not be underwater scrubbed or otherwise maintained. Data from ships subject to underwater maintenance shall not be accepted.

c. Have ship inspections conducted in accordance with ASTM F 1133. Rating of waterborne ships shall be in accordance with standard industry practices. Report: (1) checking in accordance with ASTM D 660, (2) cracking in accordance with ASTM D 661 and (3) blistering in accordance with ASTM D 714. Biofouling or physical failure due to verifiable physical damage that has removed the anticorrosive paint to the steel hull or removed the antifouling paint to the anticorrosive layers may be subtracted from the percentage area biofouled or anticorrosive paint failure.

d. Have patch tests of 30 feet in width (minimum) located amidships with underwater hull systems running from the keel to the heavy load line and boottop systems running from the heavy load line to the light load line. Patch tests shall include:

(1) A 10 foot wide (minimum) section of the complete paint system being qualified.

(2) A 10 foot (minimum) wide section of the system being qualified with the type III of the MIL-P-24441 anticorrosive system.

(3) A 10 foot (minimum) wide section of the system being qualified with the type IV of the MIL-P-24441 anticorrosive system.

(4) A 10 foot wide (minimum) section of an antifouling system already qualified to this specification.

(5) A 10 foot wide (minimum) section of a system already on the QPL of this specification.

(6) Manufacturer commercial ship patch tests shall be on commercial, Military Sealift Command (MSC), Maritime Administration (MARAD), or other government agency ships. The application and inspections shall be certified and verified by a disinterested third party mutually agreed on by NAVSEA and the paint vendor and shall be at the vendor's cost. The ship operations (travel routes, ports, times in port and lay-up times) shall be reported.

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(7) Government patch test application and inspections shall be at a naval shipyard or under Navy contract at a commercial shipyard. Application shall be verified by a Navy representative at Navy cost.

(8) Tests of paint systems intended for rubber shall be conducted only on government ships having extensive rubber surfaces.

e. Full ship applications shall consist of a full hull application of the manufacturer's total commercial system, except that the following shall be included:

(1) The antifouling paints and tie coats, if any, being qualified with the MIL-P-24441 type III and type IV anticorrosive system shall be applied as a belly band (either boottop to boottop via the keel or waterline to waterline via the keel, as appropriate) with a minimum width of 20 foot.

(2) There shall be a control paint system applied consisting or a system chosen from the QPL of this specification or as otherwise approved by the qualifying activity. This control paint system shall be applied as a belly band (either boottop to boottop via the keel or waterline to waterline via the keel, as appropriate) with a minimum width of 20 foot.

(3) Commercial, MSC, MARAD, or other government agency ships applications and inspections must be certified and verified by a disinterested third party mutually agreed on by NAVSEA and the paint vendor at the vendor's cost. The ship operations shall be certified and verifiable with travel, port time and lay-up times documented.

(4) The application and inspections of full ship applications to government ships shall be at a naval shipyard or under Navy contract at a commercial shipyard. Application and inspections shall be by a Navy representative at Navy cost.

4.5.1.3 Erosion. Erosion tests shall be prepared, installed, conducted, and inspected as required in 4.5.1.1. In addition, the tests shall be as follows.

4.5.1.3.1 Type I, all classes, all grades, applications 1 and 2 paints shall be tested for biofouling resistance for two years (12 cycles). Paint systems shall meet the requirements of 3.4.4.1.1.

4.5.1.3.2 Type I, all classes, all grades, applications 3 and 4 paints shall be tested three years (18 cycles). Inspect test samples for exposed type II, MIL-P-15931, or anticorrosion paint, as appropriate. Paint systems shall meet the requirements of 3.4.4.1.2.

4.5.1.4 Shallow submergence. shallow submergence tests shall be prepared, installed, conducted, and inspected as required in 4.5.1.1. In addition, the tests shall be as follows.

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4.5.1.4.1 Types I and II, all classes, all grades, applications 1 and 2 paints shall be tested for four years. Paint systems shall meet the requirements of 3.4.4.2.1.

4.5.1.4.2 Types I and II, all classes, all grades, applications 3 and 4 paints shall be tested for six years. Paint systems shall meet the requirements of 3.4.4.2.2.

4.5.1.5 Ship tests. Ship tests (patch or full applications) shall be installed, conducted, and inspected as required in 4.5.1.2. In addition, the tests shall be as follows.

4.5.1.5.1 Commercial ship patch test. Commercial test patch applications shall be for a period of five years. Test results shall meet the requirements of 3.4.4.3.3.

4.5.1.5.2 Commercial ship test. Commercial whole ship applications shall be for a period of five years. Test results shall meet the requirements of 3.4.4.3.4.

4.5.1.5.3 Government ship patch test. Government ship patch applications shall be for a period of two years (minimum) on a Navy ship. Test results shall meet the requirements of 3.4.4.3.5.

4.5.1.5.4 Government full ship test. Government full ship applications shall be for a period of one year (minimum) on a Navy ship. Test results shall meet the requirements of 3.4.4.3.6.

4.5.1.5.5 Service life determination. Paints in service on Navy ships will be monitored by the Government for five years or seven years, as appropriate. Paints shall meet the requirements of 3.4.4.3.7.

4.5.2 Recoatability.

4.5.2.1 Type I. Two panels from each set of test paint systems (see 4.5.1) shall be retrieved from those tested in 4.5.1.3.1 and 4.5.1.3.2, after completion of the required test. These panels shall be removed from test and shall be cleaned for a period not to exceed 2 minutes with fresh water at a nominal pressure of 750 pounds per square inch (lb/in^2). Test panels shall not be allowed to dry between being removed from test and the pressure cleaning. Dry for 24 hours at ambient conditions. Recoat one-half of each test panel with one coat of the same topcoat as originally used. Recoating shall be in accordance with the manufacturer's ASTM F 718 instructions. Application and curing of this recoat paint film shall be in accordance with the manufacturer's ASTM F 718. The recoated panels shall be tested for 3 cycles as specified in 4.5.1.3.1. Rate as specified in 4.5.1.1. Test paints shall meet the requirements of 3.4.8.

4.5.2.2 Type II. Two panels of each set of test paint systems (see 4.5.1) shall be retrieved from those tested in 4.5.1.4.1 and 4.5.1.4.2, after completion of the required test. These panels shall be removed from test and shall be cleaned for a period not to exceed 2 minutes with fresh water at a nominal pressure of 750 lb/in^2. Test panels shall not be allowed to dry

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between being removed from test and the pressure cleaning. Dry for 24 hours at ambient conditions. Recoat one-half of each test panel with one coat of the same topcoat as originally used. Recoating shall be in accordance with the manufacturer's ASTM F 718 instructions. Application and curing of this recoat paint film shall be in accordance with the manufacturer's ASTM F 718. The recoated panels shall be tested for 3 months as specified in 4.5.1.4.1. Rate as specified in 4.5.1.1. Test paints shall meet the requirements of 3.4.8.

4.5.3 Volatile organic content (VOC). VOC shall be determined in accordance with EPA 40 CFR CH. 1, part 60, appendix A, method 24. The VOC of each paint shall comply with the requirements of 3.3.3, 3.5, table I and 3.8.

4.5.4 Storage stability.

4.5.4.1 Shelf life. After storage for two years under warehouse conditions (see 4.5), the paint shall be tested for the properties required in 3.4.3, in accordance with the test listed in table II. Paints shall conform to the requirements in 3.4.3.1.

4.5.4.2 Partially full container. Store a sealed, two-thirds full container (250 milliliter [ml] or pint [pt]) of each individual paint for seven days at 140°F (60°C). Skinning shall be determined after 48 hours in accordance with method 3021.1 of FED-STD-141. After completion of the seven day storage, the paint shall be tested for the properties required in 3.4.3, as specified in the test listed in table II. Paint shall conform to the requirements in 3.4.3.2.

4.5.4.3 Accelerated storage stability. After exposure to a temperature of 140°F (60°C) (nominal) continually for 30 days, a sealed, full one-quart container of paint shall be tested for the properties required in 3.4.3, as specified in the test listed in table II. Paint shall conform to the requirements in 3.4.3.3.

4.5.5 Spraying properties. Each individual paint of the paint system shall be mixed in accordance with the manufacturer's ASTM F 718 data sheet. Using the manufacturer's ASTM F 718 designated spray equipment, spray each paint onto individual ground glass panels (see 4.5.1.1) to the manufacturer's ASTM F 718 recommended wet film thickness. The panel shall be observed for spraying properties. The dry film shall be inspected for running, sagging, or streaking, blooming, blushing, bubbling, cratering, dusting, floating, fogging, hazing, mottling, orange peel appearance, pinholing, seeding, or streaking in accordance with method 4331.1 of FED-STD-141. The test results for each individual paint shall meet the requirements of 3.4.5.

4.5.6 Brushing properties. The brushing properties of the paint shall be determined in accordance with method 4321.1 of FED-STD-141. The dry film shall be inspected for running, sagging, or streaking, blooming, blushing, bubbling, cratering, dusting, floating, fogging, hazing, mottling, orange peel appearance, pinholing, seeding, or streaking. The test results for each individual paint shall meet the requirements of 3.4.6.

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4.5.7 Rolling properties. The rolling properties of the paint in an "as received" condition shall be determined in accordance with method 4335 of FED-STD-141. The dry film shall be inspected for running, sagging, or streaking, blooming, blushing, bubbling, cratering, dusting, floating, fogging, hazing, mottling, orange peel appearance, pinholing, seeding, or streaking. The test results for each individual paint shall meet the requirements of 3.4.7.

4.5.8 Color.

4.5.8.1 Application 2 and 4. The color of application 2 and 4 antifouling topcoat paints and individual paints (except anticorrosive topcoats in R_0 colors) shall be determined in accordance with method 4250.1 of FED-STD-141A. In case of doubt, color differences (delta L, delta a and delta b) shall be determined in accordance with ASTM D 2244 using a D-65 light source, a 45 degree illumination angle and a 0 degree viewing angle. Color shall be in accordance with the requirements of 3.4.1 (A) and (B), as appropriate.

4.5.8.2 Application 1 and 3. Test panels of each R_0 dark gray color paint shall be prepared and allowed to cure at least 24 hours before testing. The ASTM D 2244 L_H value (see paragraph 6.3 of ASTM D 2244) color of the R_0 anticorrosive topcoats shall be determined in accordance with ASTM E 97. Before measuring the color, wet the surface of the painted test panel with water to which a small amount of colorless surfactant has been added (a water solution containing less than 1 percent surfactant). The area to be measured should be completely coated with the water film when it is inserted into the reflectometer for measurement. Check the test panel for complete water coverage after the measurement is completed. If the water film does not completely cover the area just measured, discard the measurement and repeat the process until a measurement is obtained on a test panel having complete water coverage both at the start and the end of the measurement. This ASTM D 2244 L_H value obtained is defined as R_w. The R_w measured shall conform to the appropriate requirements of 3.4.1 (B).

4.5.9 Freeze-thaw stability. A 1-pint, resin-lined, friction top can shall be filled two-thirds full with the paint and the can tightly closed. The closed can and its contents shall be exposed three times to the following temperature cycle:

a. Low temperature of 16 to 20°F (-9 to -7°C) for 16 hours.

b. High temperature of 77 to 82°F (25 to 28°C) for 8 hours.

At the completion of the exposure cycles, the consistency and condition in container shall be determined, in accordance with ASTM D 562 and method 3011.2 of FED-STD-141. Consistency shall be compared with the original consistency to determine compliance with requirements in 3.4.10.2. The condition in container shall meet the requirements of 3.4.10.2.

4.5.10 Cathodic protection compatibility. Two steel panels from each set of test paint systems (see 4.5.1) shall be retrieved from those tested in 4.5.1.1 after completion of the required test. Each test panel shall be electrically connected by an insulated copper wire to a MIL-A-21412 magnesium

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anode and shall have a 1/8-inch (nominal) hole drilled through the coating to the metal at the center of the test panel. The electrical resistance between a point on the surface of the anode and the metal in the drilled hole of the test panel shall be less than 0.01 ohm, when checked with an ohm meter. Connecting points on the test panel shall be coated with an epoxy compound for insulation. The test panel shall be reinstalled in the ASTM D 3623 test in such a manner as to separate the test panel from the magnesium anode by two feet (nominal) for a period of 3 months. At the completion of the three-month test, inspect each test panel for lifting or undercutting around the drilled hole, peeling, blistering, dissolving, or other paint film defects. Test results shall be in accordance with the requirements of 3.4.9.

4.5.11 Soluble and total metal content. Soluble and total metal content shall be determined on a dry paint film mode from each individual paint of the paint system in accordance with the Federal Register (FR), Volume 55, paragraph 11798, March 19, 1990 (55 FR 11798), Toxicity Characteristic Leaching Procedure (TCLP). Test results shall be in accordance with the requirements of 3.3.5.

4.6 Paints and coatings product/procedure data. Data sheets shall be in accordance with the requirements of 3.3.5.2, and 3.5. Manufacturer's commercial and ASTM F 718 data sheets shall reflect the information required in 3.2 for application purposes. The toxic(s) (pesticides, biocides, and so forth) used in the product being qualified shall be in accordance with the requirements of the class for which the product is being qualified (see 3.3.2 and 6.4).

4.7 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.8 Inspection of packaging. The packaging, packing, marking and data sheets of the paint shall be examined to determine compliance with the requirements of Section 5 of this specification.

5. PACKAGING

5.1 Packaging, packing and marking. The paint shall be packaged level A, B or C, packed level A, B or C, as specified (see 6.2), and marked in accordance with PPP-P-1892. Marking shall include hazardous warnings in accordance with MIL-STD-129 (see 4.7 and 6.2), FED-STD-313, and 29 CFR Parts 1910, 1915, 1917, 1918, 1926 and 1928, as applicable. Unless otherwise specified (see 6.2), the paint shall be furnished in one- or five-gallon pails (or metric equivalent) in accordance with PPP-P-404, type II, class three with wire handles or bails treated to resist corrosion. Materials used in packaging and packing shall conform to PPP-F-320, PPP-P-1892, PPP-T-60, and MIL-L-19140. An MSDS and ASTM F 718 data sheet shall be included with all packaged or packed paint containers (see 3.5, 3.7 and 6.5).

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5.1.1 Special marking. Each unit container shall have the following information:

a. Specification number.

b. Type, class, grade and application.

c. Color.

d. Manufacturer's name, address, product commercial nomenclature designation, and QPL number.

e. Hazardous materials information required by MIL-STD-129 (see 4.7 and 6.2), FED-STD-313, and 29 CFR Parts 1910, 1915, 1917, 1918, 1926, and 1928, as applicable.

f. Required storage information including temperatures.

g. Antifouling paints shall have the EPA approved label on each container. Containers of antifouling topcoat paints may have the marking information required in (a) and (f) above in printed materials affixed to the container in a weather-proof pouch to comply with EPA requirements.

5.1.1.1 Containers shall also be marked as follows:

"Contains a maximum of INSERT PROPER NUMBER HERE pounds per gallon (INSERT PROPER NUMBER HERE grams per liter) of volatile organic content (VOC) per 40 CFR CH. 1, Part 60, Appendix A, (U. S. EPA) method 24."

"This product may be thinned (INSERT QUANTITY and CONTAINER SIZE HERE) per five gallon container with IDENTIFY GENERIC AND TRADEMARK THINNER IDENTIFICATIONS HERE and will contain less than (INSERT VOC REGULATORY NUMBERS LEVEL HERE) volatile organic content (VOC) per 40 CFR CH. 1, Part 60, Appendix A, (U. S. EPA) method 24."

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. This specification covers a variety of high grade, paint systems. These coatings are intended for application to ship hulls to prevent marine biofouling and corrosion. The type I antifouling topcoat paints are intended to polish (erode or ablate) under water flow conditions to provide a biofouling free, smooth hull for fuel efficiency. Type II antifouling topcoat paints are intended to provide biofouling control without eroding. Paints are provided in seven classes of toxic agents suitable for steel, aluminum, and rubber surfaces. Antifouling paint conforming to present air protection regulations (1990) is required as grade A, but specification

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provides grades at lesser VOC levels to encourage manufacturers to develop more environmentally acceptable materials.

Except for the toxic agent containing antifouling topcoats of classes 1A, 1B, 2, 3A and 3B, the individual dried paint residues and dry paint contaminated debris may constitute non-hazardous waste.

6.1.1 The following figure, which outlines the biological resistance testing requirements of this specification, is for information only. Manufacturers are advised to review the requirements and quality sections for the actual requirements.

	Time in years *					
TEST	0	1	2	3	4	5
EROSION PANEL TEST (3.4.4.1)	S----------C					
STATIC PANEL TEST (3.4.4.2)	S----------C					
COMMERCIAL SHIP PATCH TEST (3.4.4.3.3)	S----------C					
NAVY SHIP PATCH TEST (3.4.4.3.5)			S----------C			
COMMERCIAL WHOLE SHIP APPLICATION (3.4.4.3.4)	S----------C					
WHOLE NAVY SHIP APPLICATION (3.4.4.3.6)					S---C	
SERVICE LIFE TESTING			1/			

* S (START)----C (COMPLETE) defines interval of required test.

1/ Test period of 5 or 7 years depending on application (see 3.3.4)

FIGURE 1. Biological resistance performance testing requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Type, class, grade, and application of paint required (see 1.2).

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. Certifications (if any) required (see 3.3.5.2).

e. Required marking (see 5.1).

f. Level of packaging and of packing required (see 5.1).

g. Size of container required, if other than as specified (see 5.1).

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h. Special markings required (see 5.1.1).

i. Data sheets (see 6.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-L exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.3.5.2	DI-E-2121	Certificate of Compliance	---
4.4	DI-MISC-80678	Certification/data report	10.3.1

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List No. 24647 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, D. C. 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4.1).

6.4.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.5 Material Safety Date Sheets and ASTM F 718 data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313 and ASTM F 718 data sheets. The pertinent Government mailing addresses for submission of data are in FED-STD-313.

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6.6 Subject term (key word) listing.

Ablative
Copper
Copper oxide
Erodible
Organotin
VOC

6.7 Changes to previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH

Review activity:
Army - MR

User Activity:
Navy - CG
Navy - YD

Preparing activity:
Navy - SH
(Project 8010-0405)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-24647B	2 April 1991

3. DOCUMENT TITLE
PAINT SYSTEM, ANTICORROSIVE AND ANTIFOULING, SHIP HULL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact (TPOC) Dr. Brenda Holmes (SEA 5141) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS	TPOC: 703 602-0215	8 332-0215
c. ADDRESS (Include Zip Code) Commander, Naval Sea System Command Department of the Navy (SEA 55Z332) Washington, D.C. 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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